



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



CVE-2026-76460 Cisco ISE Privileged API Vulnerability Daily Update

CVE / Vulnerability Threat Intelligence Report

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TLP:CLEAR | Lost Boys Cyber | CVE-2026-76460 Cisco ISE Privileged API Vulnerability Daily Update - CVE / Vulnerability Threat Intelligence Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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1. Executive Summary

CISA added CVE-2026-76460 to the KEV catalog on 2026-09-16. For defenders, the priority is accelerated exposure review for Cisco ISE deployments because identity and network-access-control systems sit on privileged trust paths into enterprise, campus, and aviation operations networks.

This daily update is intentionally dated 2026-09-20 so current GitHub-visible artifacts exist for today's defense prioritization. No claim of same-day disclosure is made; the driver is recent KEV activity and continued defender urgency.

2. Vulnerability Metadata

Field	Value
CVE	CVE-2026-76460
Product / Component	Cisco Identity Services Engine (ISE)
Weakness	Incorrect use of privileged APIs enabling unauthorized privileged action paths
Severity	Critical
Affected Versions	Validate against vendor advisory / asset inventory
Fixed Versions	Apply Cisco fixed software per PSIRT / emergency change guidance
Disclosure Date	See vendor/CVE record
Exploitation Status	Listed by CISA KEV; treat as known exploited

3. Source Review & Confidence

Source	Contribution	Confidence
CISA KEV alert	https://www.cisa.gov/news-events/alerts/2026/09/16/cisa-adds-two-known-exploited-vulnerabilities-catalog	High/Medium
CISA KEV catalog	https://www.cisa.gov/known-exploited-vulnerabilities-catalog	High/Medium
NVD CVE record	https://nvd.nist.gov/vuln/detail/CVE-2026-76460	High/Medium
Cisco security advisories portal	https://sec.cloudapps.cisco.com/security/center/publicationListing.x	High/Medium

The source base supports a high-confidence defensive priority: identify exposed assets, apply vendor fixes, and monitor likely exploitation or post-compromise pivots. Exploit mechanics should not be inferred beyond vendor/CVE/CISA language.

4. Vulnerability Details

CVE-2026-76460 affects Cisco Identity Services Engine (ISE). The available public priority signal is the KEV listing; therefore this report focuses on exposure management and detection opportunities rather than speculative proof-of-concept mechanics.

5. Attack Scenarios

- An attacker targets an unpatched deployment reachable from an administrative, VPN, or partner network and attempts privileged actions.
- A compromised administrator workstation or remote-access path is used to reach the vulnerable component and move into identity or network-control workflows.
- In aviation/aerospace environments, third-party IT, airport shared services, or MRO support paths may expose the system indirectly even when it is not internet-facing.

6. Threat Landscape & Exploitation Status

CISA KEV inclusion indicates known exploitation and requires expedited remediation for U.S. federal civilian agencies; it is also a strong prioritization signal for private-sector defenders.

7. Affected Products and Exposure

Exposure Pattern	Why It Matters
Cisco ISE administrative portals and APIs reachable from enterprise admin networks	Requires owner validation, patch SLA, and compensating monitoring.
Network-access-control policy nodes integrated with identity stores	Requires owner validation, patch SLA, and compensating monitoring.
Managed-service or shared campus deployments used by airports, MRO sites, aerospace factories, and defense suppliers	Requires owner validation, patch SLA, and compensating monitoring.

8. Mitigation and Workarounds

Priority	Owner	Action
Critical	Vulnerability management	Confirm product/version ownership and emergency patch plan.
High	Identity / network team	Restrict management access to hardened admin networks and require MFA.
High	Detection engineering	Create hunts for exploitation attempts and post-compromise account activity.
Medium	GRC / vendor management	Verify MSP and third-party management exposure.

9. Detection Engineering

9.1. Detection Logic Summary

Signal	Telemetry Source	Analytic Goal
Access to vulnerable management endpoint from unusual subnet	Web/proxy/VPN logs	Detect exposure probing or exploitation setup
Administrative changes soon after suspicious access	Identity/admin audit logs	Surface post-exploitation configuration activity

9.2. Sigma / Detection Rule

title: Suspicious Administrative Access Around CVE-2026-76460

```

status: experimental
description: Correlates unusual management access with administrative change activity for KEV-tracked
exposure response.
logsource:
  category: webserver
detection:
  selection_path:
    cs-uri-query|contains:
      - 'admin'
      - 'api'
  selection_status:
    sc-status:
      - 200
      - 401
      - 403
      - 500
  condition: selection_path and selection_status
level: high

```

9.3. Hunt Query

```

DeviceNetworkEvents
| where Timestamp > ago(14d)
| where RemoteUrl has_any ('ise', 'admin', 'api', 'pixel', 'android') or RemotePort in (443, 8443,
9443)
| summarize Connections=count(), Devices=dcount(DeviceName) by RemoteUrl, RemoteIP,
InitiatingProcessAccountName
| order by Connections desc

```

10. Threat Hunting

10.1. Hunt Hypothesis

If exploitation or targeted follow-on activity occurred, defenders may see unusual administrative access followed by new sessions, privileged changes, or abnormal device/network management activity.

10.2. Hunt Query

```

SigninLogs
| where TimeGenerated > ago(14d)
| where ResultType == 0
| summarize FirstSeen=min(TimeGenerated), LastSeen=max(TimeGenerated), Apps=make_set(AppDisplayName,
20), IPs=make_set(IPAddress, 20) by UserPrincipalName
| where array_length(IPs) > 3 or array_length(Apps) > 5

```

11. MITRE ATT&CK Mapping

Technique	Name	Evidence / Rationale
T1190	Exploit Public-Facing Application	Applicable where vulnerable services are reachable by attackers.
T1078	Valid Accounts	Post-compromise or adjacent identity abuse is a plausible defender watchpoint.
T1021	Remote Services	Administrative access paths may be used for follow-on movement.

12. Validation / Lab Testing

No exploit reproduction was performed in this automated daily run. Validation completed: source review, KEV/CVE prioritization, report render, PDF text extraction, and detection-query preservation checks.

13. Recommended Actions Summary

Priority	Owner	Action
Critical	Asset owner	Patch or isolate confirmed vulnerable assets.
High	SOC	Hunt for suspicious access and privileged changes for the last 30 days.
High	IAM	Review admin accounts and enforce phishing-resistant MFA.
Medium	Business continuity	Confirm manual fallback for identity/network-control dependencies.

14. References

- [1] CISA KEV alert: <https://www.cisa.gov/news-events/alerts/2026/09/16/cisa-adds-two-known-exploited-vulnerabilities-catalog>
- [2] CISA KEV catalog: <https://www.cisa.gov/known-exploited-vulnerabilities-catalog>
- [3] NVD CVE record: <https://nvd.nist.gov/vuln/detail/CVE-2026-76460>
- [4] Cisco security advisories portal: <https://sec.cloudapps.cisco.com/security/center/publicationListing.x>